



I2200: Digital Image processing

Lecture 4: Spatial Domain Processing and Image Enhancement



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Outline:

- Intensity Transformation
 - Intensity transformation functions (negative, log, gamma), intensity and bit-place slicing, contrast stretching
 - Histograms: equalization, matching, local Processing
- Spatial Filtering
 - Filtering basics, smoothing filters, sharpening filters, unsharp masking, Laplacian
- Combining spatial operations

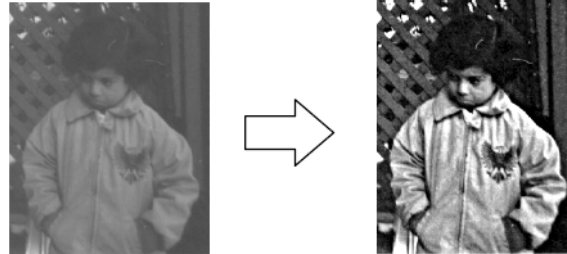


Spatial Domain Processing

- **Spatial domain processing:** Methods based on direct manipulation on pixels in an image.
- **Enhancement:** the process of manipulating an image so that the result is more suitable than the original for a specific application.

Spatial Domain Processing

- Application



- Method

$$f \xrightarrow{T_N(.)} g = T_N(f)$$

$$f(x, y) \quad , \quad 1 \leq x \leq M, 1 \leq y \leq N$$

$$g(x, y) \quad , \quad 1 \leq x \leq M, 1 \leq y \leq N$$

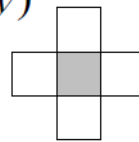
$T_N(.)$: Spatial operator defined on a neighborhood N of a given pixel

$N_0(x, y)$

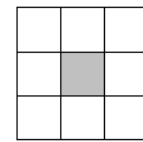


point processing

$N_4(x, y)$



$N_8(x, y)$



**mask/kernel
processing**

Intensity transformation

- Map a given gray or color level u to a new level v

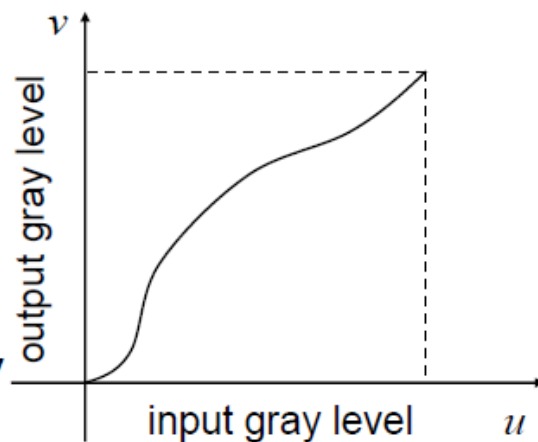
$$f(x, y) \rightarrow g(x, y)$$

$$v = \mathcal{T}(u)$$

$$x = 1, \dots, M, \quad y = 1, \dots, N$$

$$u, v = 0, \dots, 255$$

- Memory-less, direction-less operation
 - output at (x, y) only depend on the input intensity at the same point
 - Pixels of the same intensity gets the same transformation
- Does not bring in new information, may cause loss of information
- But can improve visual appearance or make features easier to detect



Intensity transformation example

■ Color Space Transformation



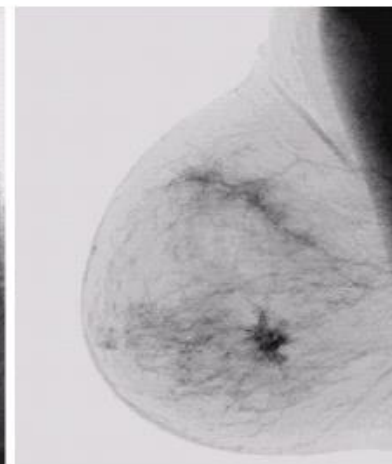
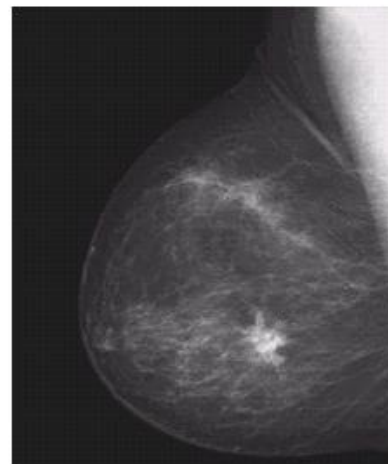
Image negatives

$$v = 255 - u \quad u, v = 0, \dots, 255$$



the appearance of
photographic
negatives

- Enhance white or gray detail on dark regions, esp. when black areas are dominant in size



a b
FIGURE 3.4
(a) Original digital mammogram.
(b) Negative image obtained using the negative transformation in Eq. (3.2-1).
(Courtesy of G.E. Medical Systems.)

Basic intensity transform functions

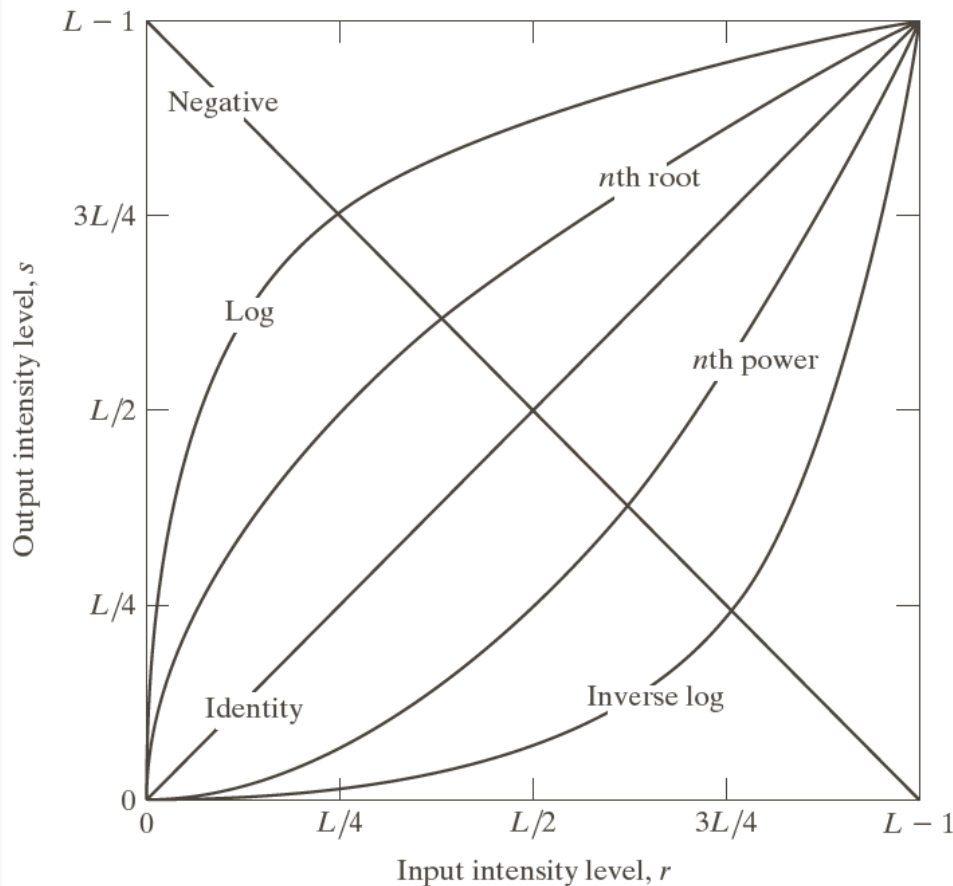


FIGURE 3.3 Some basic intensity transformation functions. All curves were scaled to fit in the range shown.

- monotonic, reversible
- compress or stretch certain range of gray-levels



Log transform

- The general form of the log transformation is

$$s = c * \log(1 + r)$$

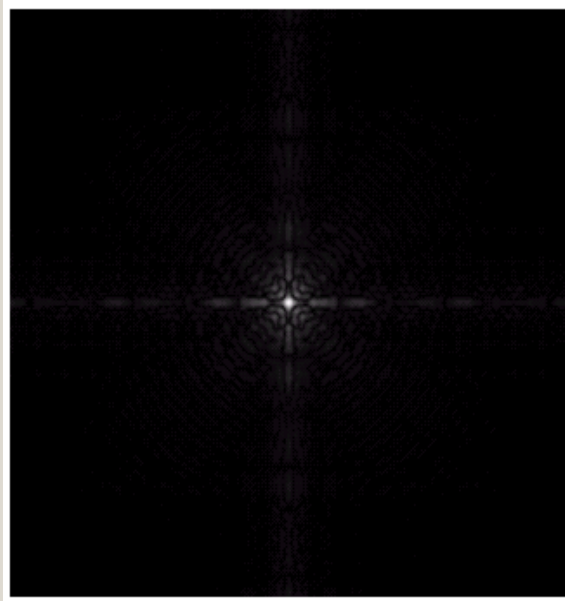
- The log transformation maps a narrow range of low input grey level values into a wider range of output values
- The inverse log transformation performs the opposite transformation

We usually set c to 1

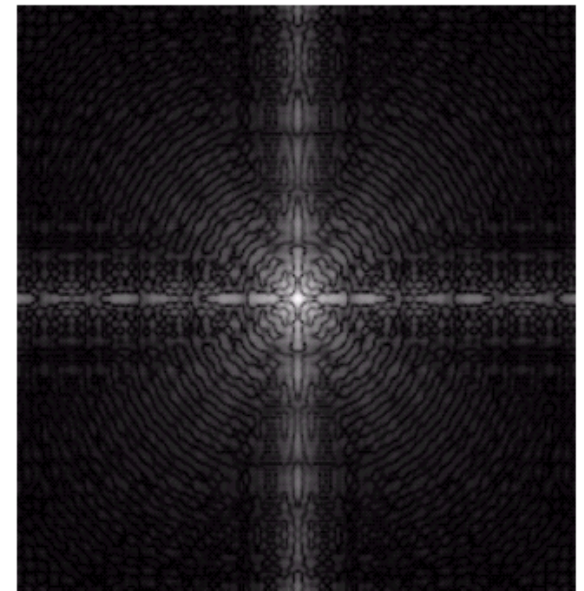
Grey levels must be in the range $[0.0, 1.0]$

Log transform example

- In the following example the Fourier transform of an image is put through a log transform to reveal more detail



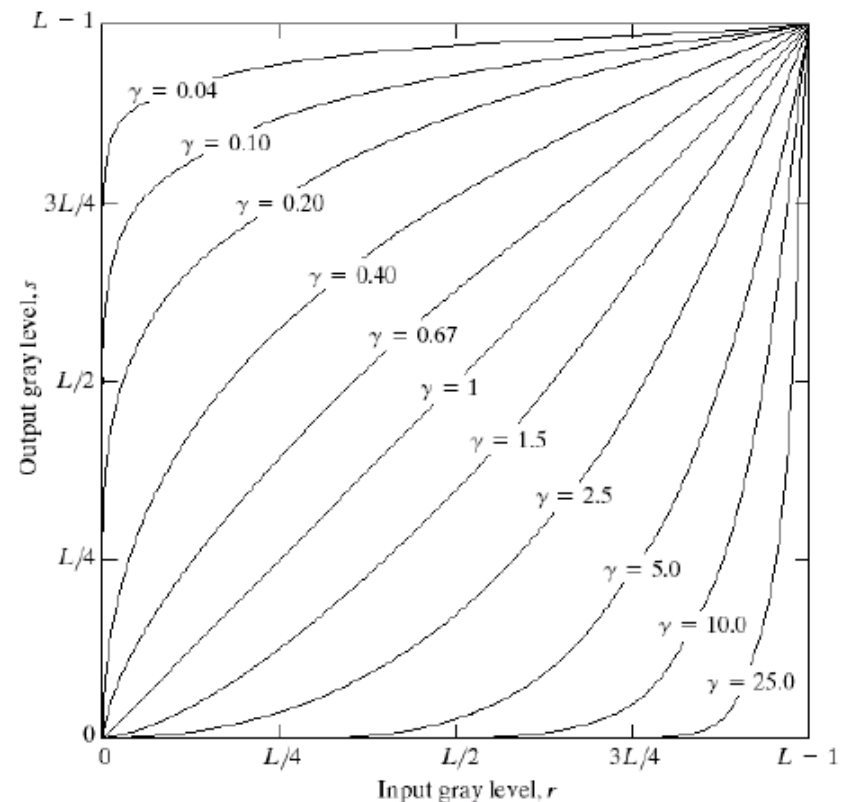
$$s = \log(1 + r)$$



Power-law (Gamma) transformation

$$v = c \cdot u^\gamma$$

- power-law response functions in practice
 - CRT Intensity-to-voltage function has $\gamma \approx 1.8 \sim 2.5$
 - Camera capturing distortion with $\gamma_c = 1.0 \sim 1.7$
 - Similar device curves in scanners, printers, ...



- power-law transformations are also useful for general purpose contrast manipulation

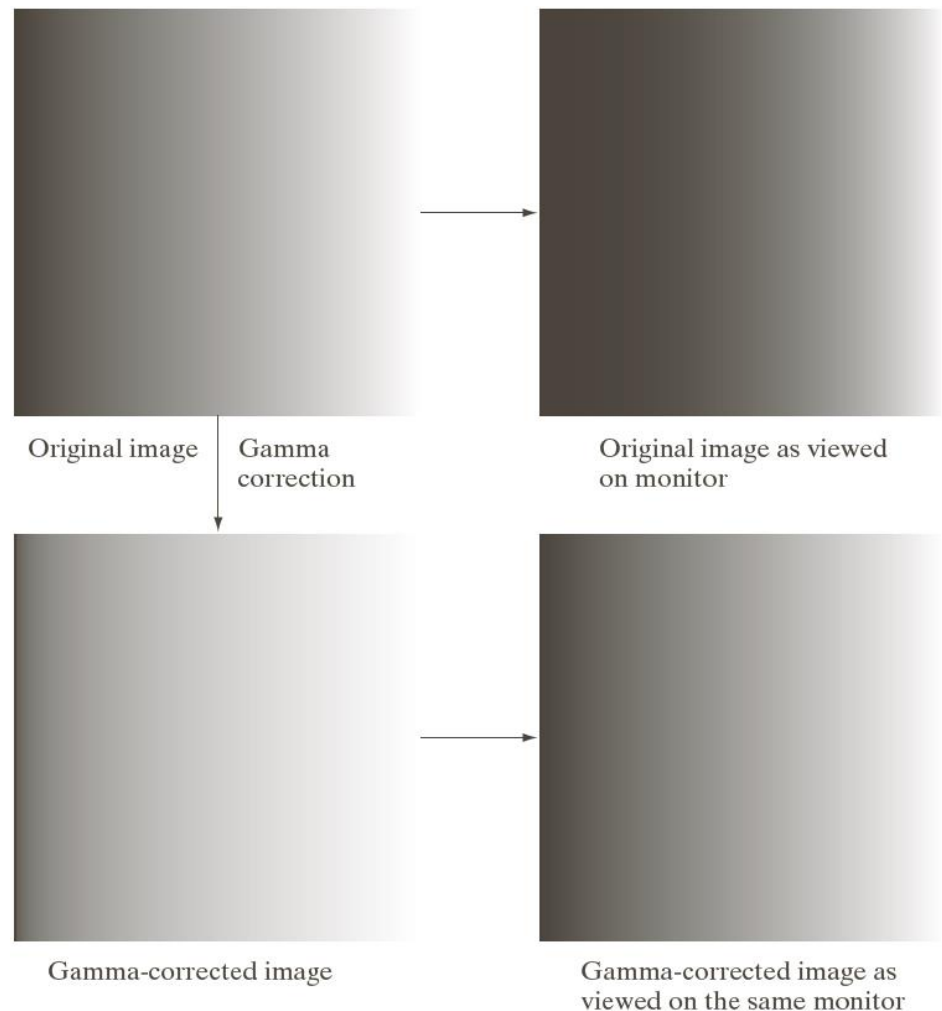
Gamma correction

- Make linear input appear linear on displays
- Method: calibration pattern + interactive adjustment



FIGURE 3.7

(a) Intensity ramp image. (b) Image as viewed on a simulated monitor with a gamma of 2.5. (c) Gamma-corrected image. (d) Corrected image as viewed on the same monitor. Compare (d) and (a).



Effect of gamma on consumer photo

$L_0^{2.2}$



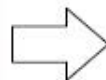
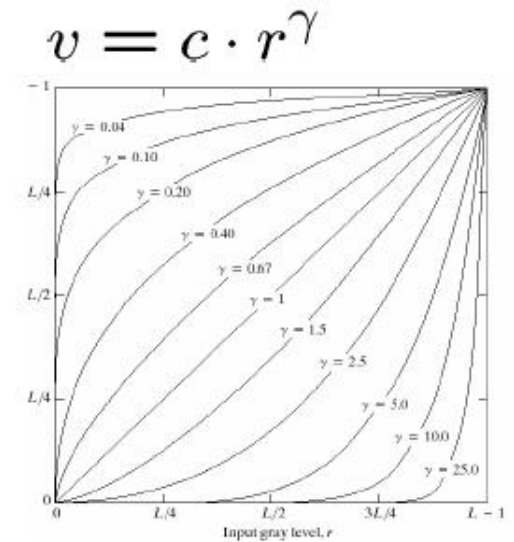
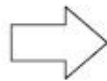
L_0



$L_0^{1/2.2}$



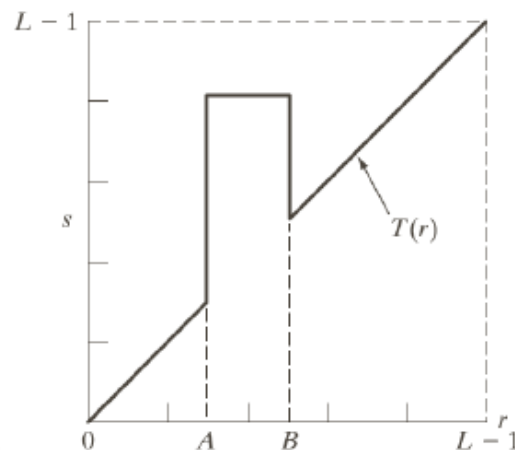
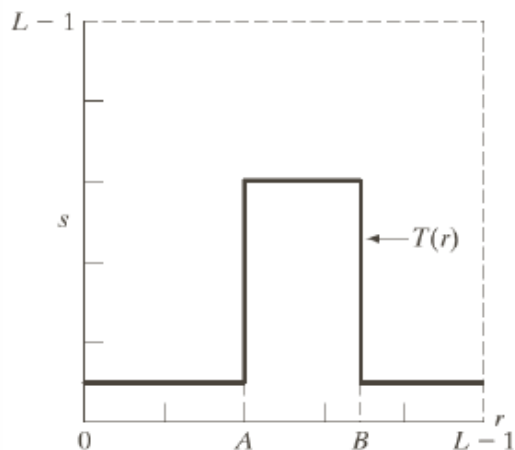
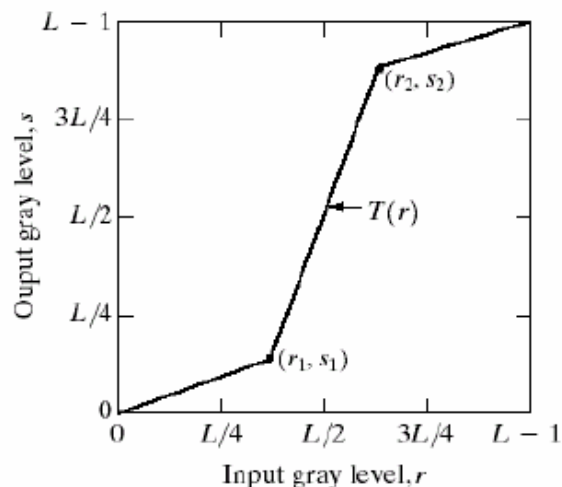
What gamma to use?



$\gamma > 1$
 $\gamma < 1$?

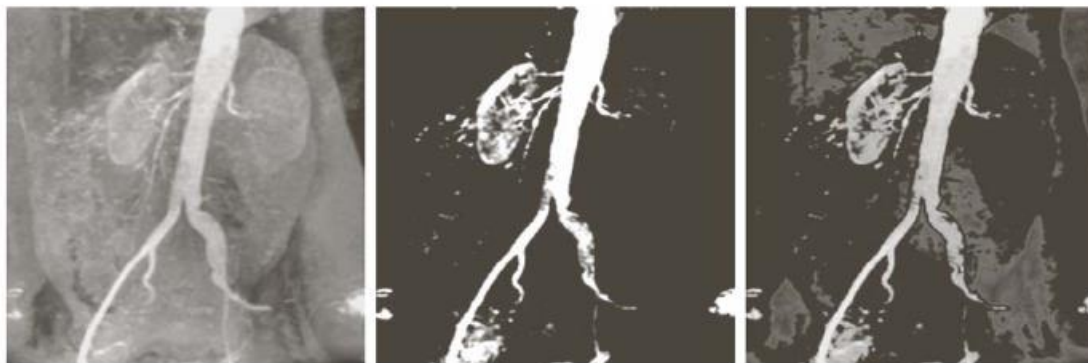
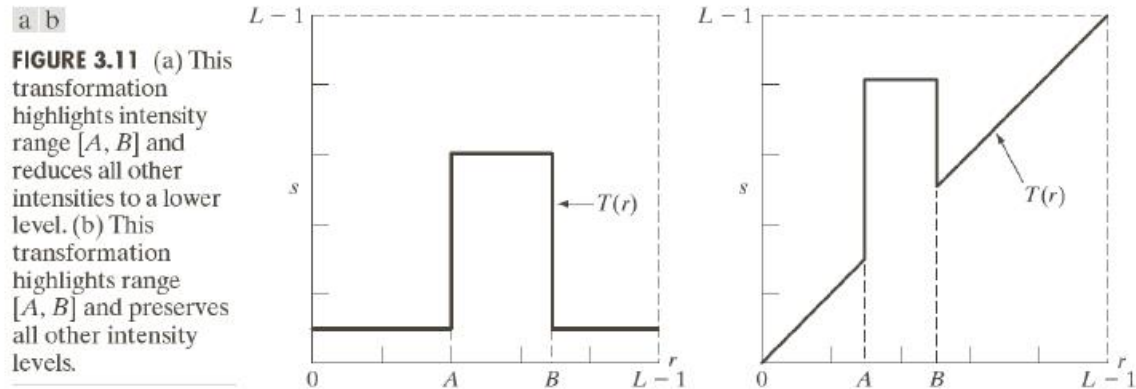
More intensity transform

- log, gamma ... closed-form functions on $[0,1]$
- can be more flexible



contrast stretching

Intensity slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Image bit-planes

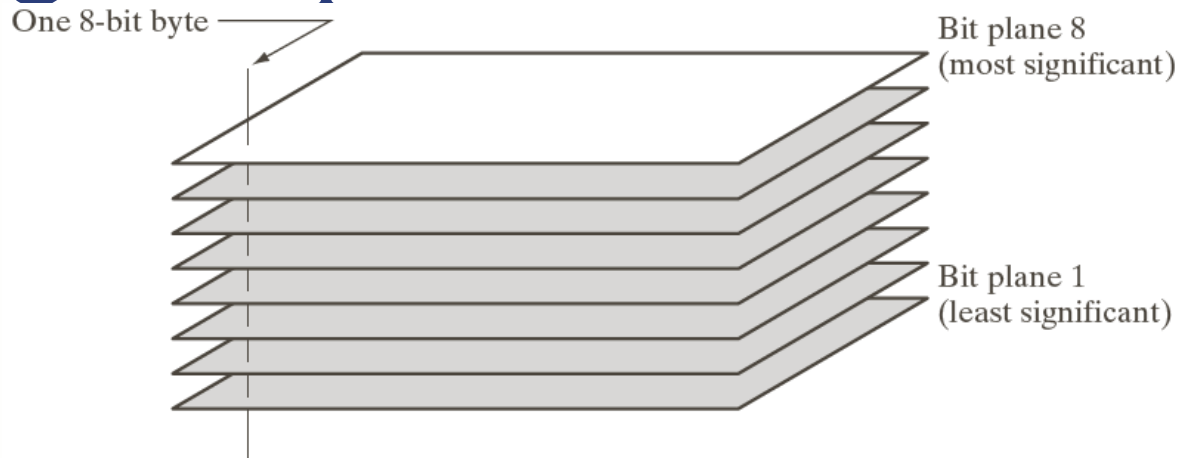


FIGURE 3.13
Bit-plane
representation of
an 8-bit image.



a	b	c
d	e	f
g	h	i

FIGURE 3.14 (a) An 8-bit gray-scale image of size 500×1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

Slicing bit-planes

- Depend on relative importance of bits
- How much to slice depend on image content
- Useful in image compression, e.g. JPEG2000



a b c

FIGURE 3.15 Images reconstructed using (a) bit planes 8 and 7; (b) bit planes 8, 7, and 6; and (c) bit planes 8, 7, 6, and 5. Compare (c) with Fig. 3.14(a).



Image bit-planes

For an 8-bit intensity value 181, the binary representation is 10110101.

Bit plane	Value	Contribution	Cumulative total
1st	1	$1 \times 2^7 = 128$	128
2nd	0	$0 \times 2^6 = 0$	128
3rd	1	$1 \times 2^5 = 32$	160
4th	1	$1 \times 2^4 = 16$	176
5th	0	$0 \times 2^3 = 0$	176
6th	1	$1 \times 2^2 = 4$	180
7th	0	$0 \times 2^1 = 0$	180
8th	1	$1 \times 2^0 = 1$	181

https://en.wikipedia.org/wiki/Bit_plane



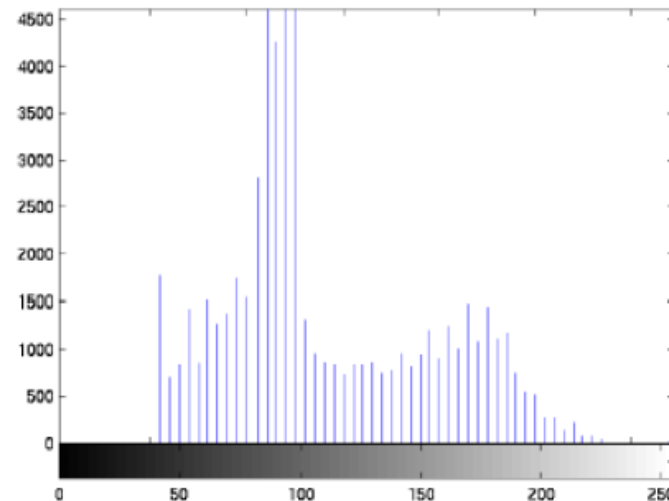
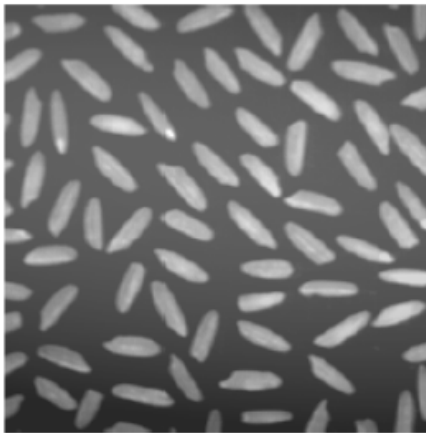
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Gray-level image histogram

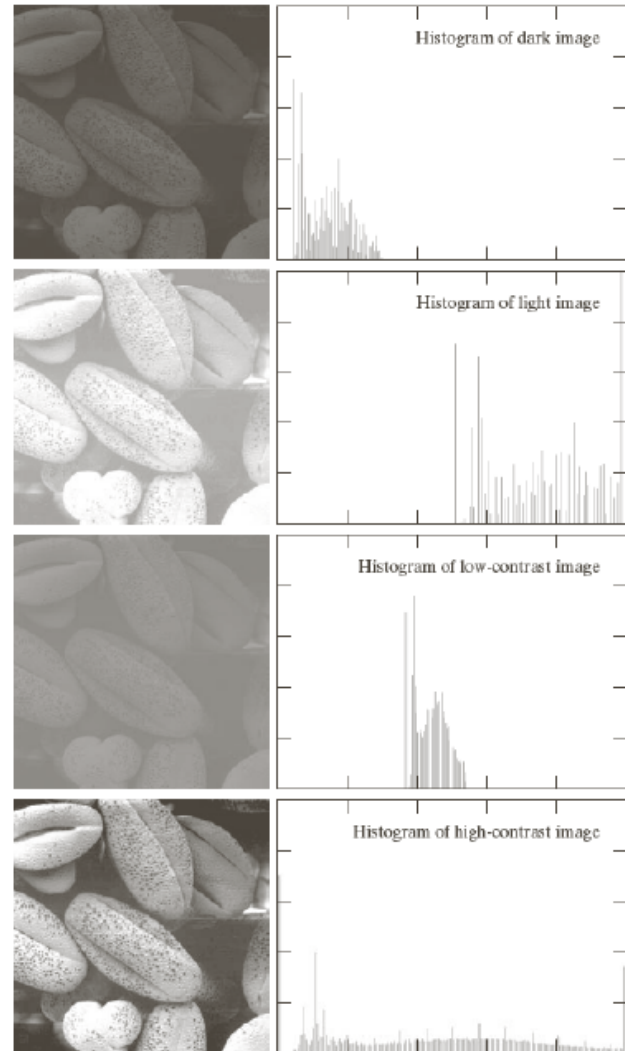
- Represents the relative frequency of occurrence of the various gray levels in the image
 - For each gray level, count the number of pixels having that level
 - Can group nearby levels to form a big bin & count #pixels in it

```
I = imread('rice.tif');  
imshow(I)  
figure, imhist(I,64)
```

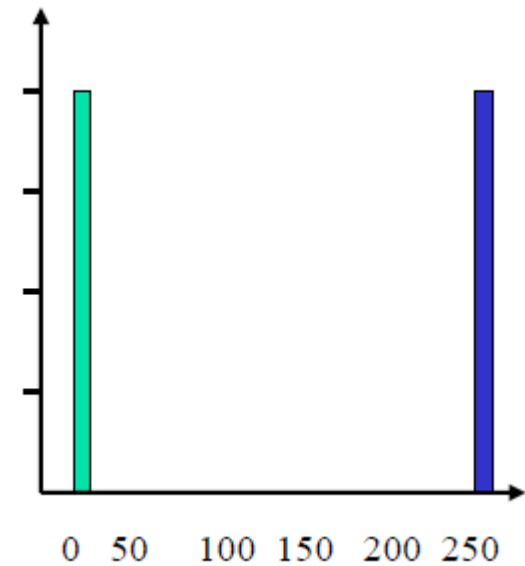
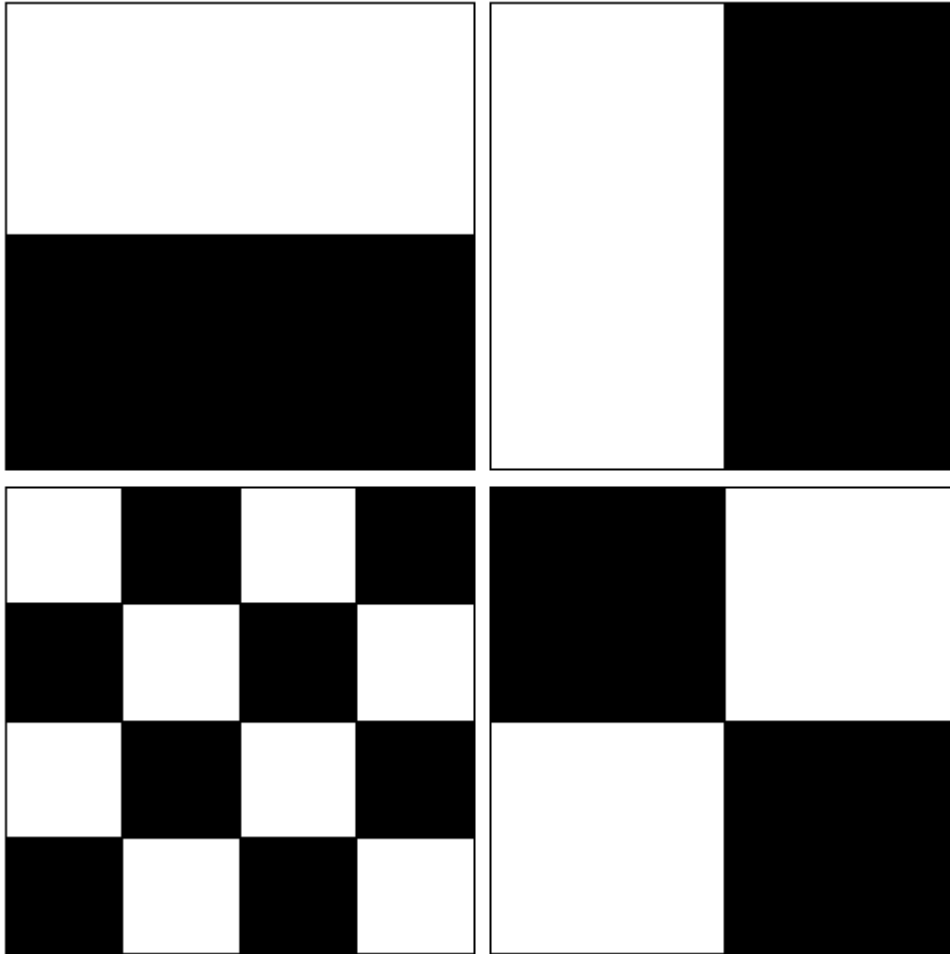


Interpretations of histogram

- if pixel values are i.i.d random variables $\rightarrow$ histogram is an estimate of the probability distribution of the r.v.
- “unbalanced” histograms do not fully utilize the dynamic range
 - Low contrast image: narrow luminance range
 - Under-exposed image: concentrating on the dark side
 - Over-exposed image: concentrating on the bright side
- “balanced” histogram gives more pleasant look and reveals rich details



Very Different Images with Same Histograms

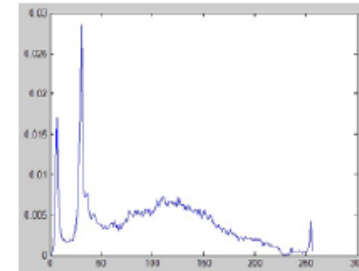
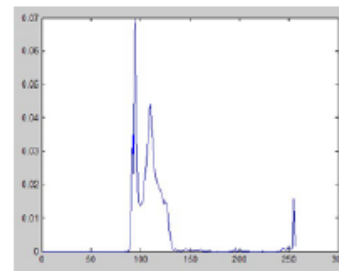
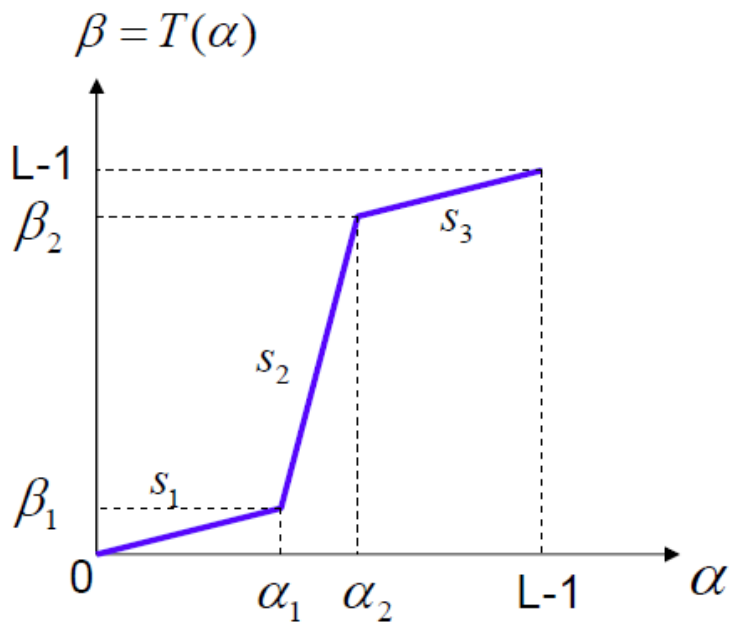


Contrast stretching

Stretch the over-concentrated gray-levels

Piece-wise linear function, where the slope in the stretching region is greater than 1.

- $I_{output} = (I - I_{min}) * 255 / (I_{max} - I_{min})$



The Probability Density Function of an Image

Let
$$A = \sum_{g=0}^{255} h_{I_k}(g+1).$$

$h_I(g+1)$ = the number
of pixels in I
with graylevel g .

Note that since $h_{I_k}(g+1)$ is the number of pixels in I_k (the k th color band of image I) with value g , A is the number of pixels in I . That is if I is R rows by C columns then $A = R \times C$.

pdf
[lower case]

Then,

$$p_{I_k}(g+1) = \frac{1}{A} h_{I_k}(g+1)$$

This is the probability
that an arbitrary pixel
from I_k has value g .

is the graylevel probability density function of I_k .



The Probability Density Function of an Image

- $p_{\text{band}}(g+1)$ is the fraction of pixels in (a specific band of) an image that have intensity value g .
- $p_{\text{band}}(g+1)$ is the probability that a pixel randomly selected from the given band has intensity value g .
- Whereas the sum of the histogram $h_{\text{band}}(g+1)$ over all g from 1 to 256 is equal to the number of pixels in the image, the sum of $p_{\text{band}}(g+1)$ over all g is 1.
- p_{band} is the **normalized histogram** of the band.

The Probability Density Function of an Image

Let $\mathbf{q} = [q_1 \ q_2 \ q_3] = I(r, c)$ be the value of a randomly selected pixel from I . Let g be a specific graylevel. The probability that $q_k \leq g$ is given by

PDF
[upper case]

$$P_{I_k}(g+1) = \sum_{\gamma=0}^g p_{I_k}(\gamma+1) = \frac{1}{A} \sum_{\gamma=0}^g h_{I_k}(\gamma+1) = \frac{\sum_{\gamma=0}^g h_{I_k}(\gamma+1)}{\sum_{\gamma=0}^{255} h_{I_k}(\gamma+1)},$$

where $h_{I_k}(\gamma+1)$ is the histogram of the k th band of I .

This is the probability that any given pixel from I_k has value less than or equal to g .

The Probability Density Function of an Image

Let $\mathbf{q} = [q_1 \ q_2 \ q_3] = I(r, c)$ be the value of a randomly selected pixel from I . Let g be a specific graylevel. The probability that $q_k \leq g$ is given by

Also called CDF for "Cumulative Distribution Function".

$$P_{I_k}(g+1) = \sum_{\gamma=0}^g p_{I_k}(\gamma+1) = \frac{1}{A} \sum_{\gamma=0}^g h_{I_k}(\gamma+1) = \frac{\sum_{\gamma=0}^g h_{I_k}(\gamma+1)}{\sum_{\gamma=0}^{255} h_{I_k}(\gamma+1)},$$

where $h_{I_k}(\gamma+1)$ is the histogram of the k th band of I .

This is the probability that any given pixel from I_k has value less than or equal to g .



The Probability Density Function of an Image

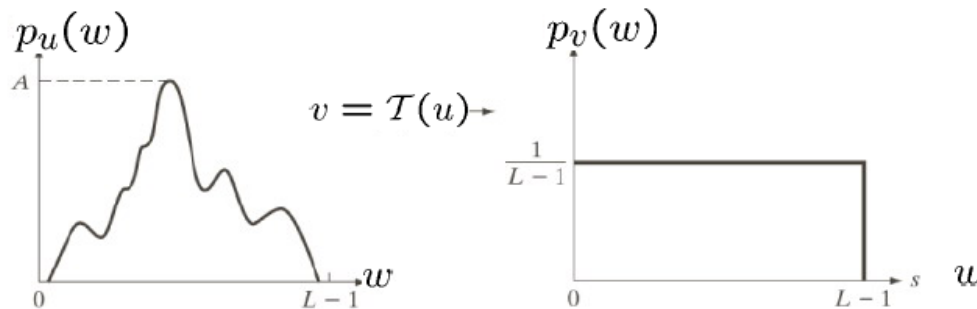
- $P_{\text{band}}(g+1)$ is the fraction of pixels in (a specific band of) an image that have intensity values less than or equal to g .
- $P_{\text{band}}(g+1)$ is the probability that a pixel randomly selected from the given band has an intensity value less than or equal to g .
- $P_{\text{band}}(g+1)$ is the cumulative (or running) sum of $p_{\text{band}}(g+1)$ from 0 through g inclusive.
- $P_{\text{band}}(1) = p_{\text{band}}(1)$ and $P_{\text{band}}(256) = 1$; $P_{\text{band}}(g+1)$ is nondecreasing.

Note: the Probability Distribution Function (PDF, capital letters) and the Cumulative Distribution Function (CDF) are exactly the same things. Both PDF and CDF will refer to it. However, pdf (small letters) is the *density* function.

Histogram equalization

- goal: map the each luminance level to a new value such that the output image has approximately uniform distribution of gray levels
- two desired properties
 - monotonic (non-decreasing) function: no value reversals
 - $[0,1] \rightarrow [0,1]$: the output range being the same as the input range

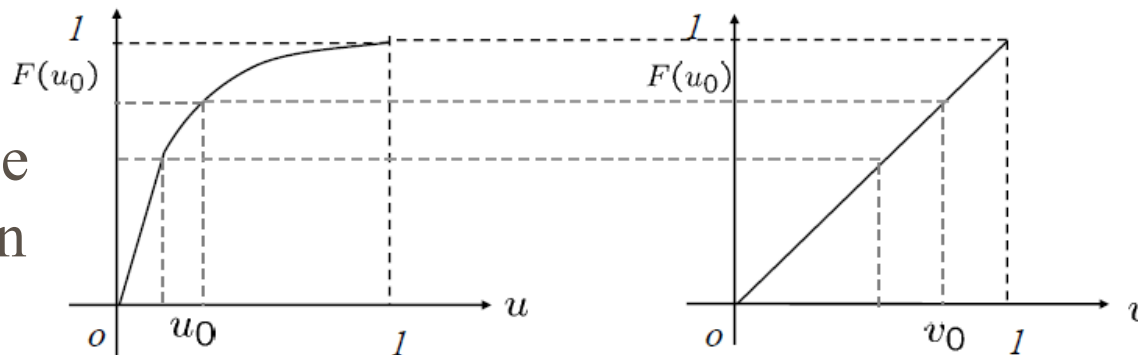
pdf
Probability
density
function



$$F_u(U) = P(U < u) = \int_0^u p_u(w)dw$$

$$F_v(V) = P(V < v) = \int_0^v p_v(w)dw$$

cdf
Cumulative
distribution
function



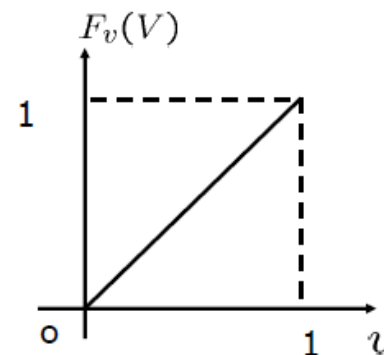
Histogram equalization

- make

$$v = F_u(u) = P(U < u) = \int_0^u p_u(w)dw$$

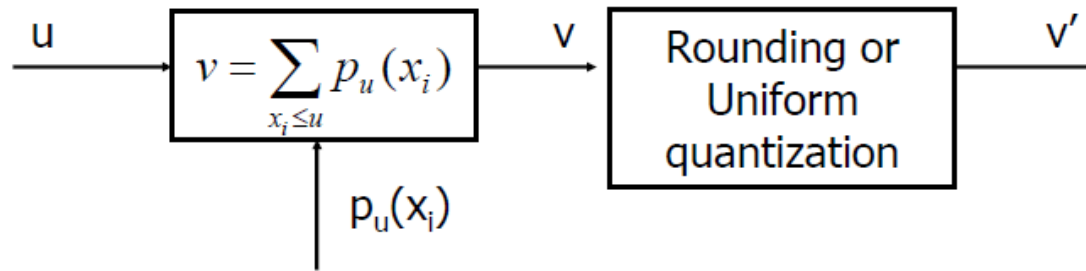
- show that v follows uniform distribution, i.e.

$$F_v(V) = v, \quad v \in [0, 1]$$



$$\begin{aligned} F_v(V) &= P(V < v) = P(F_u(U) < v) \\ &= P(U < F_u^{-1}(v)) = F_u(F_u^{-1}(v)) = v \end{aligned}$$

Implementing histogram equalization



compute histogram

$$p_u(x_i) = \frac{n(x_i)}{\sum_{i=0}^{L-1} n(x_i)} \text{ for } i = 0, \dots, L-1$$

equalize

$$v = (L-1) \sum_{x_i=0}^u p_u(x_i)$$

or

$$v = \frac{L-1}{MN} \sum_{x_i=0}^u n(x_i)$$

round the output

$$v' = \text{round}(v)$$

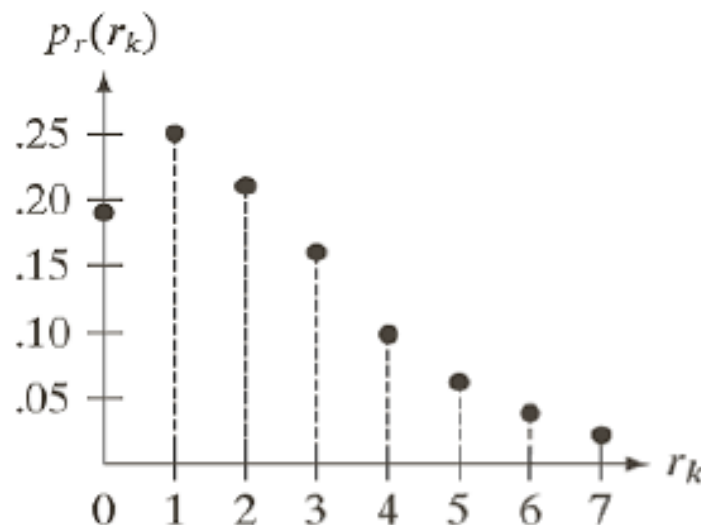
- Only depend on the input image histogram
- Fast to implement
- For u in discrete prob. distribution, the output v will be approximately uniform

Histogram equalization example

$$v = (L-1) \sum_{x_i=0}^u p_u(x_i)$$

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

$$7 * \sum_u p(u) \quad v$$



Histogram equalization example

r_k	n_k	$p_r(r_k) = n_k/MN$	$7 * p(u)$	$7 * \sum_u p(u)$	v
$r_0 = 0$	790	0.19	1.33	1.33	1
$r_1 = 1$	1023	0.25	1.75	3.08	3
$r_2 = 2$	850	0.21	1.47	4.55	5
$r_3 = 3$	656	0.16	1.02	5.67	6
$r_4 = 4$	329	0.08	0.56	6.23	6
$r_5 = 5$	245	0.06	0.42	6.65	7
$r_6 = 6$	122	0.03	0.21	6.86	7
$r_7 = 7$	81	0.02	0.14	7.00	7

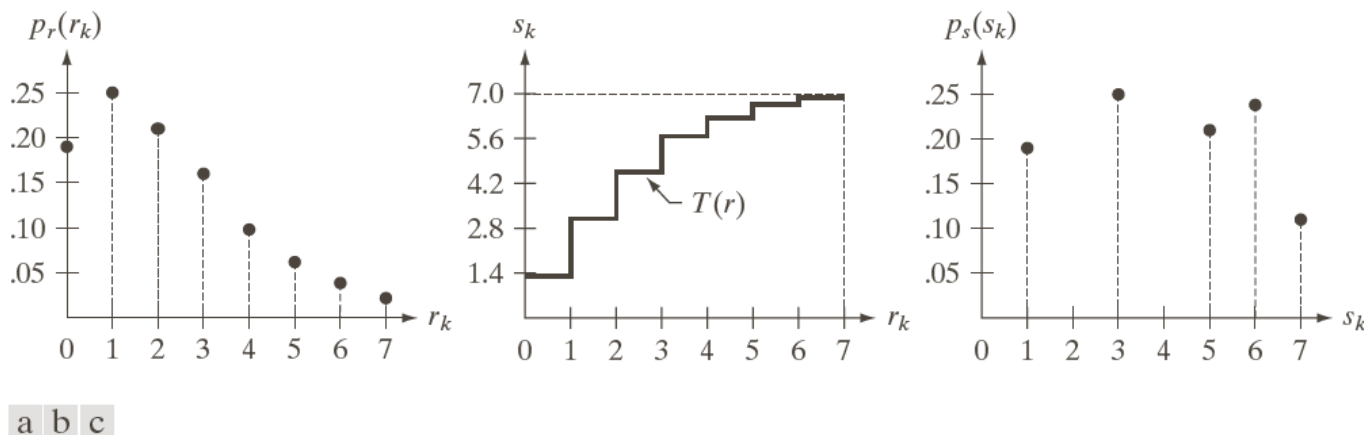


FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Histogram equalization example

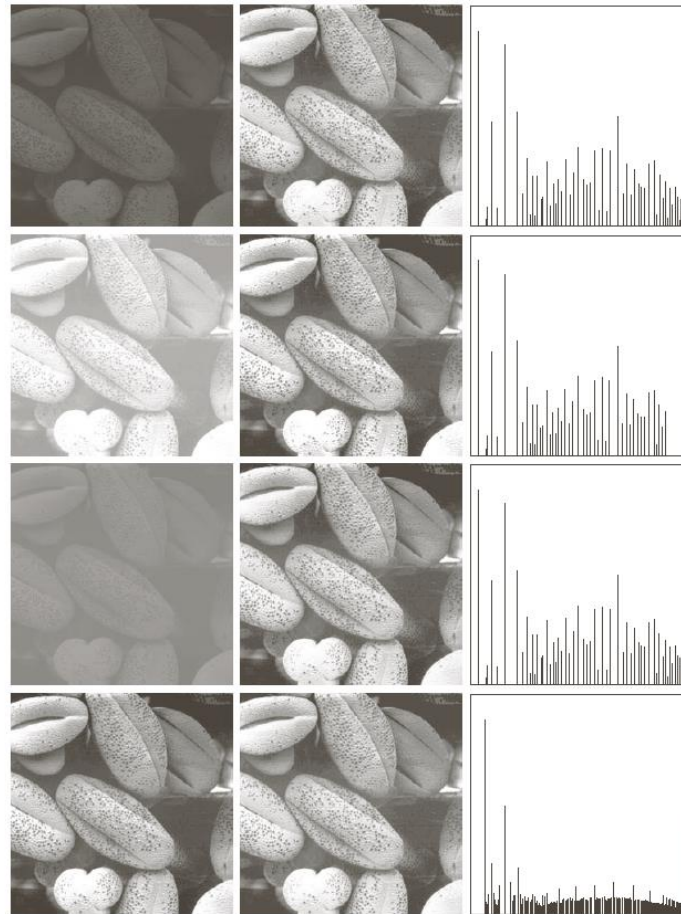
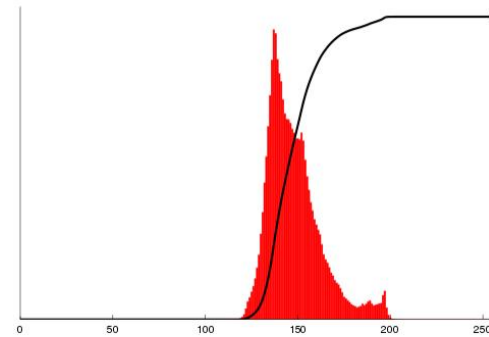


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

Histogram equalization example



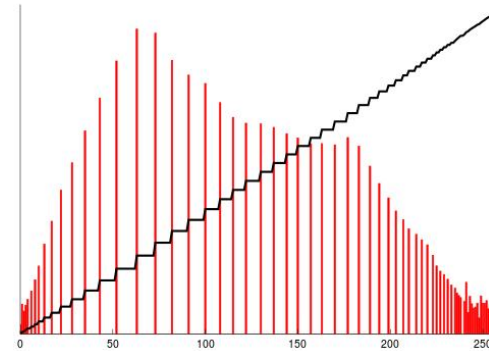
Before Histogram Equalization



Corresponding histogram (red) and cumulative histogram (black)



After Histogram Equalization



Corresponding histogram (red) and cumulative histogram (black)





Histogram equalization for color image

- ❑ Applying histogram equalization separately to the Red, Green and Blue channels – may cause **color imbalance**.
- ❑ Converting RGB color image to HSI, YUV,..., then applying histogram equalization to only the “**Intensity**” channel.

Local histogram processing – Image enhancement

- **Problem:** global spatial processing not always desirable
- **Solution:** apply point-operations to a pixel neighborhood with a sliding window

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

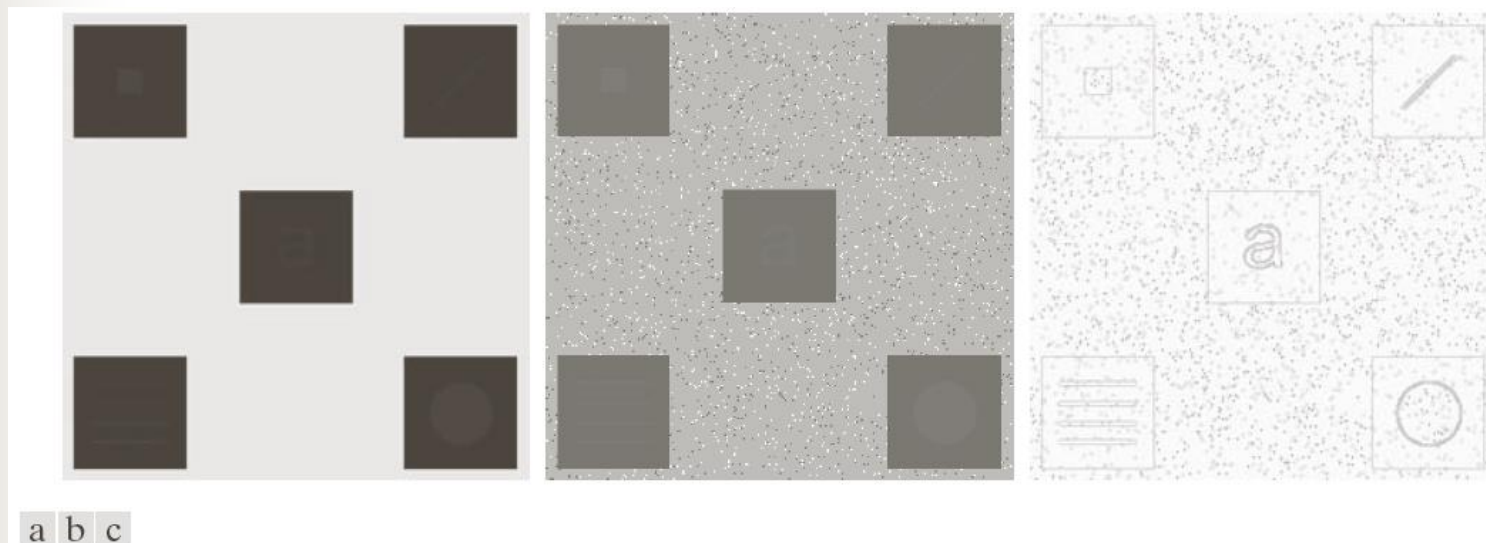


FIGURE 3.26 (a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization applied to (a), using a neighborhood of size 3×3 .



Contrast Stretching vs Histogram equalization

- Contrast stretching -- improve the contrast by 'stretching' the range of intensity values to the full range of pixel values. It is a linear stretching.
- It will **NOT** loss information.

$$I_{out} = 255 \frac{(I_{in} - I_{min})}{(I_{max} - I_{min})}$$



Contrast Stretching vs Histogram equalization

- Histogram equalization employs a monotonic, non-linear mapping which re-assigns the intensity values of pixels in the input image such that the output image contains a uniform distribution of intensities (*i.e.* a flat histogram).
- It will loss information.

Contrast Stretching vs Histogram equalization



Original

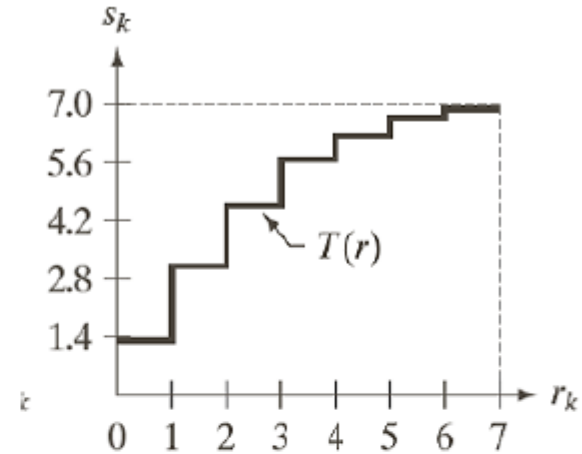
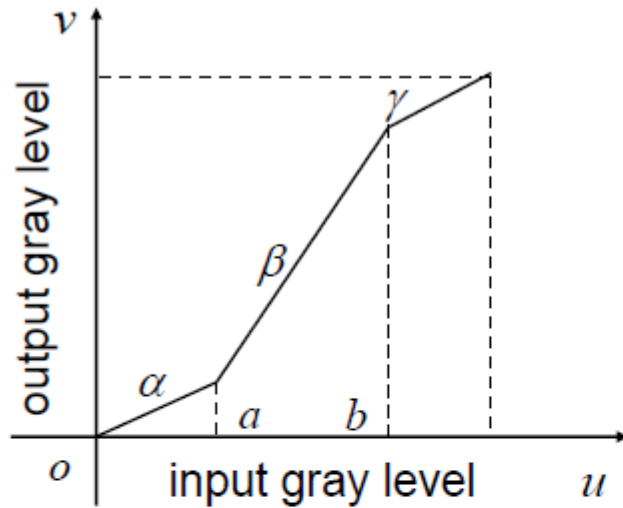


contrast stretching
 $I_{min} = 79$ and
 $I_{max} = 136$



Histogram
equalization

Contrast-stretching vs. histogram equalization



- function form
- reversible? loss of information?
- input/output?



5-Min Break



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Spatial filtering in image neighborhoods

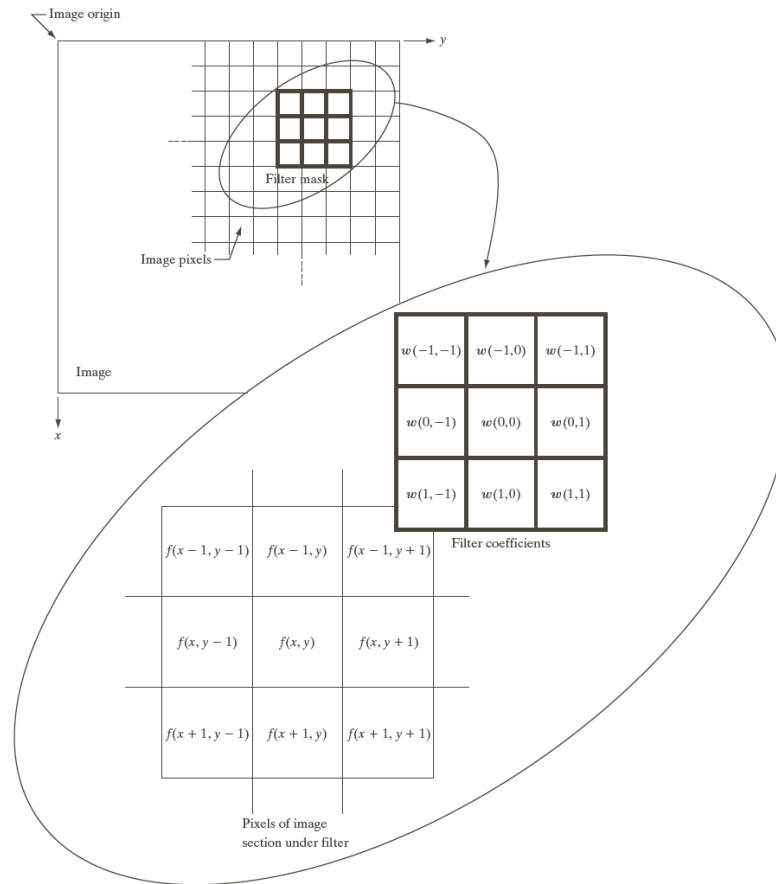
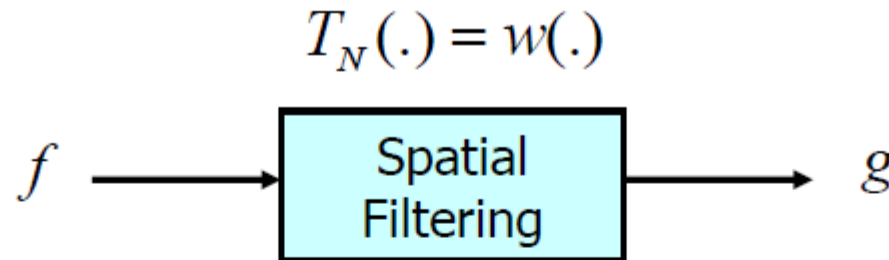


FIGURE 3.28 The mechanics of linear spatial filtering using a 3×3 filter mask. The form chosen to denote the coordinates of the filter mask coefficients simplifies writing expressions for linear filtering.

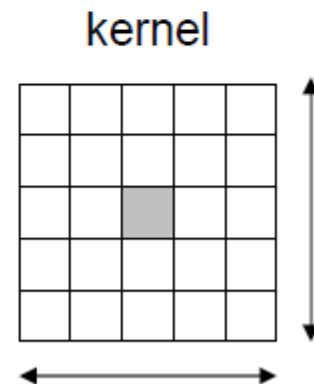
Kernel operator / filter masks



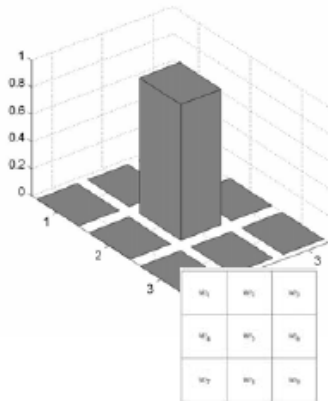
$$g(m, n) = \sum_{i=-a}^a \sum_{j=-b}^b w(i, j) f(m+i, n+j)$$

$$1 \leq m \leq M$$

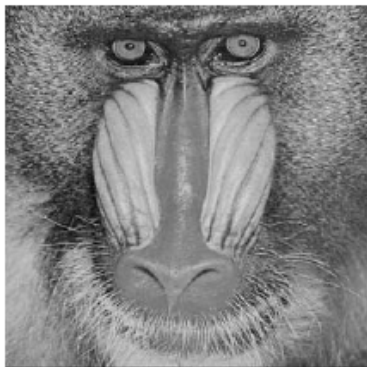
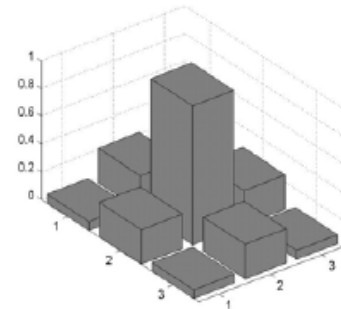
$$1 \leq n \leq N$$



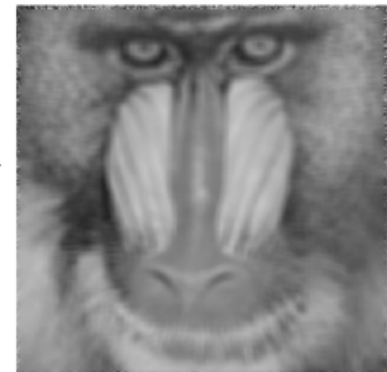
Smoothing: image averaging



$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$



smoothing
operator



Low-pass filter, leads to
softened edges

Spatial averaging can suppress noise

- image with iid noise $y(m,n) = x(m,n) + N(m,n)$
- averaging
$$v(m,n) = (1/N_w) \sum x(m-k, n-l) + (1/N_w) \sum N(m-k, n-l)$$
 - N_w : number of pixels in the averaging window
 - Noise variance reduced by a factor of N_w
 - SNR improved by a factor of N_w
 - Window size is limited to avoid excessive blurring

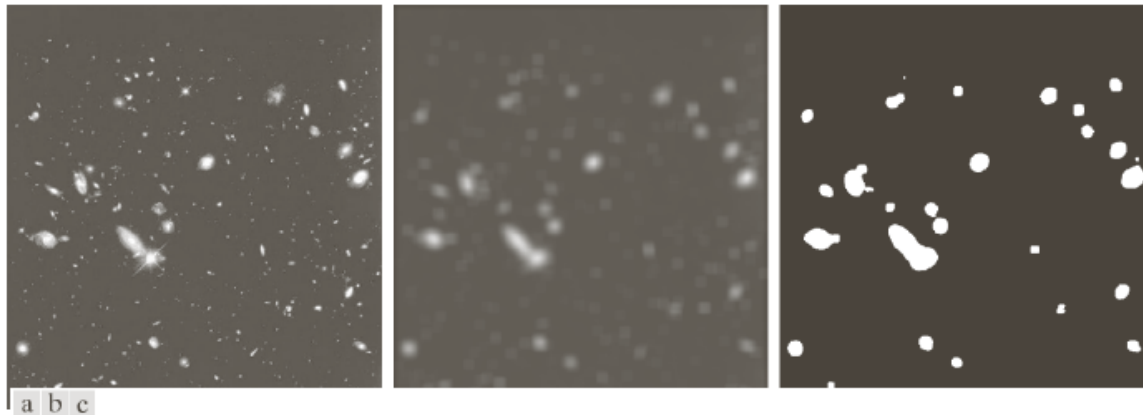


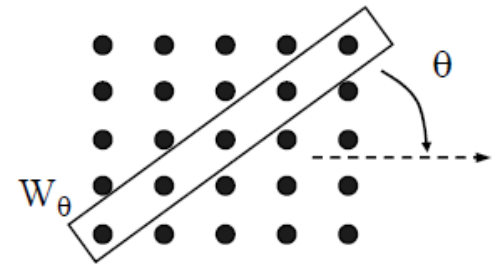
FIGURE 3.34 (a) Image of size 528×485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Smoothing operator of different sizes



Directional smoothing

- Problems with simple spatial averaging mask
 - Edges get blurred
- Improvement
 - Restrict smoothing to along edge direction
 - Avoid filtering across edges
- Directional smoothing
 - Compute spatial average along several directions
 - Take the result from the direction giving the smallest changes before & after filtering
- Other solutions
 - Use more explicit edge detection and adapt filtering accordingly



Non-linear smoothing operator

- Median filtering

- median value ξ over a small window of size N_w

$$\tilde{x} = \text{sort}(x); \quad \xi = \tilde{x}\left[\frac{N_w + 1}{2}\right]$$

- nonlinear

- $\text{median}\{x(m) + y(m)\} \neq \text{median}\{x(m)\} + \text{median}\{y(m)\}$

- odd window size is commonly used

- 3x3, 5x5, 7x7

- 5-pixel “+”-shaped window

- for even-sized windows take the average of two middle values as output

- Other order statistics: min, max, x-percentile ...

Median filter example

- Median filtering
 - resilient to statistical outliers
 - incurs less blurring
 - simple to implement

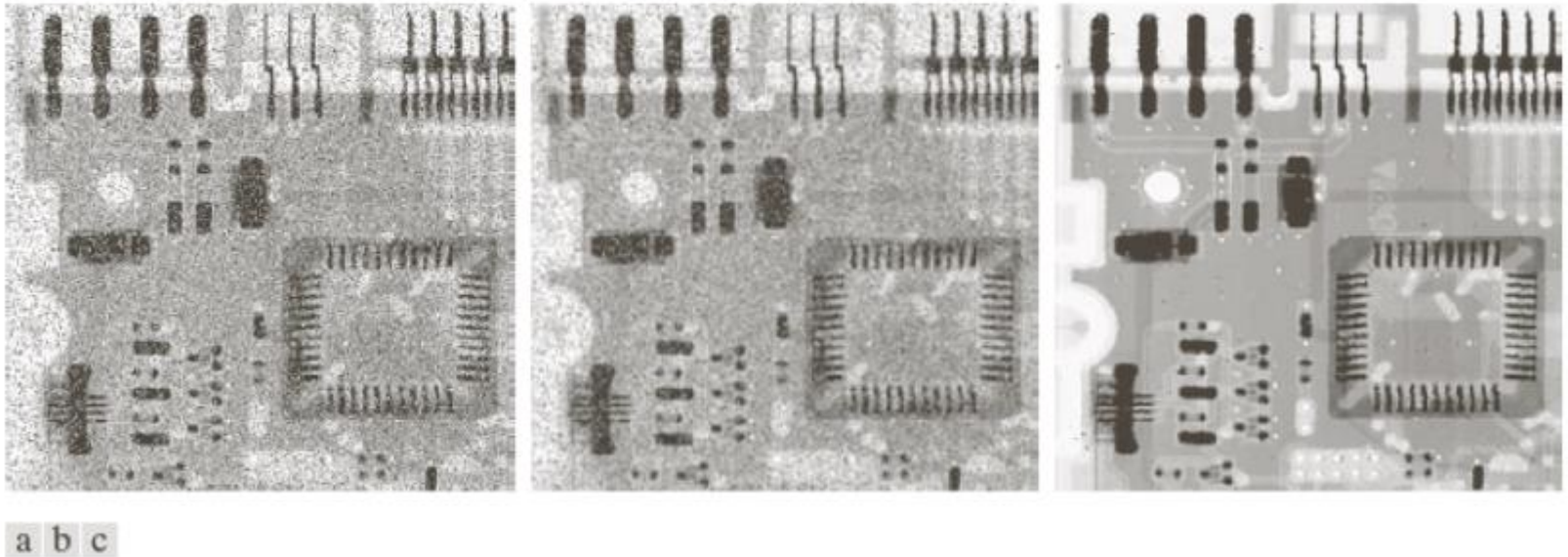


FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Image derivative and sharpening

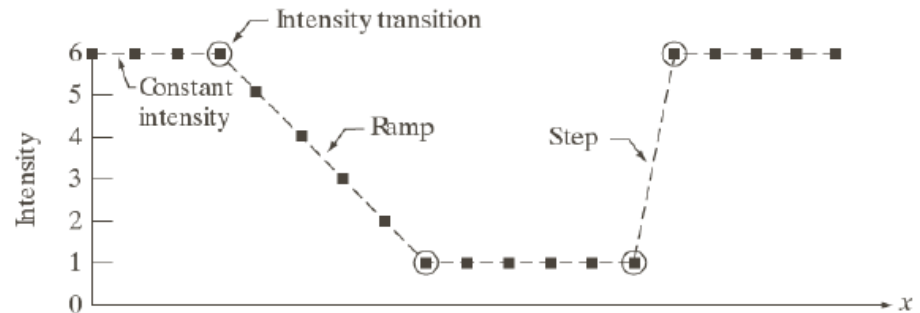
$$f'(x) = \frac{\partial f}{\partial x}$$

$$= f(x+1) - f(x)$$

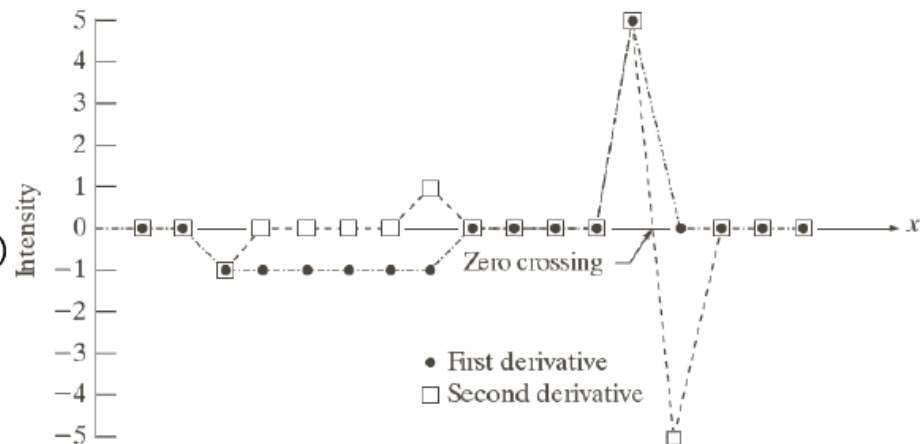
$$f''(x) = \frac{\partial^2 f}{\partial x^2}$$

$$= \frac{\partial f}{\partial x}(f'(x) - f'(x-1))$$

$$= f(x+1) + f(x-1) - 2f(x)$$

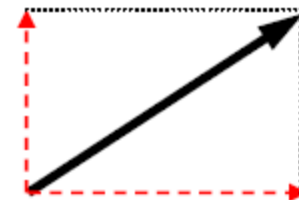
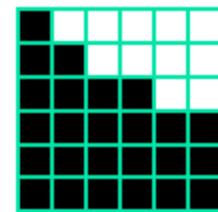


Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
1st derivative	0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0
2nd derivative	0	0	0	-1	0	0	0	0	1	0	0	0	0	0	5	-5	0	0	0



Edge and the first derivative

- Edge: pixel locations of abrupt luminance change
- Spatial luminance gradient vector
 - a vector consists of partial derivatives along two orthogonal directions
 - gradient gives the direction with highest rate of luminance changes
- Representing edge: edge intensity + directions
- Detection Methods
 - prepare edge examples (templates) of different intensities and directions, then find the best match
 - measure transitions along 2 orthogonal directions



Edge detection operators

Image gradient:

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\|\nabla f\| \approx |G_x| + |G_y|$$

Robert's
operator

Sobel's
operator

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	0	0	-1
0	1	1	0

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

a
b c
d e

FIGURE 3.41

A 3×3 region of an image (the z s are intensity values). (b)–(c) Roberts cross gradient operators. (d)–(e) Sobel operators. All the mask coefficients sum to zero, as expected of a derivative operator.

Edge detection examples



Roberts



Sobel



<http://flickr.com/photos/reneemarie11/97326485>
<http://flickr.com/photos/valkyrie2112/2829322895>

Second derivative in 2D

Image Laplacian: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\Rightarrow \nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1
0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a b
c d

FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4). (b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

Laplacian of roman ruins

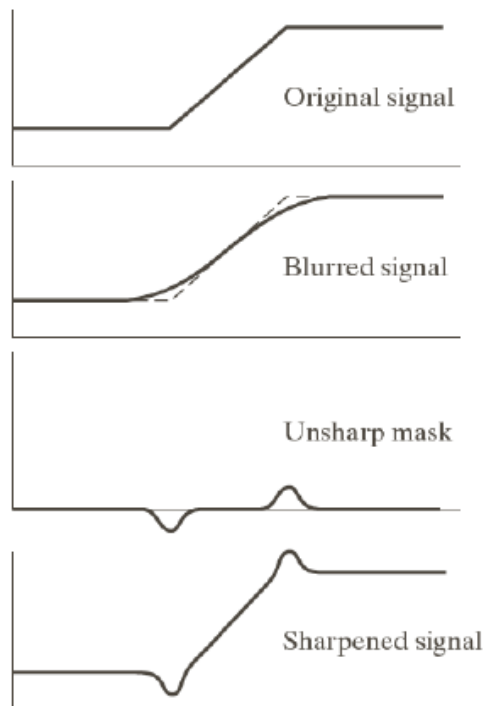


Unsharp masking

- **Unsharp masking** is an image manipulation technique for increasing the apparent sharpness of photographic images.
- The "unsharp" of the name derives from the fact that the technique uses a blurred, or "unsharp", positive to create a "mask" of the original image. The unsharp mask is then combined with the negative, creating a resulting image sharper than the original.

a
b
c
d

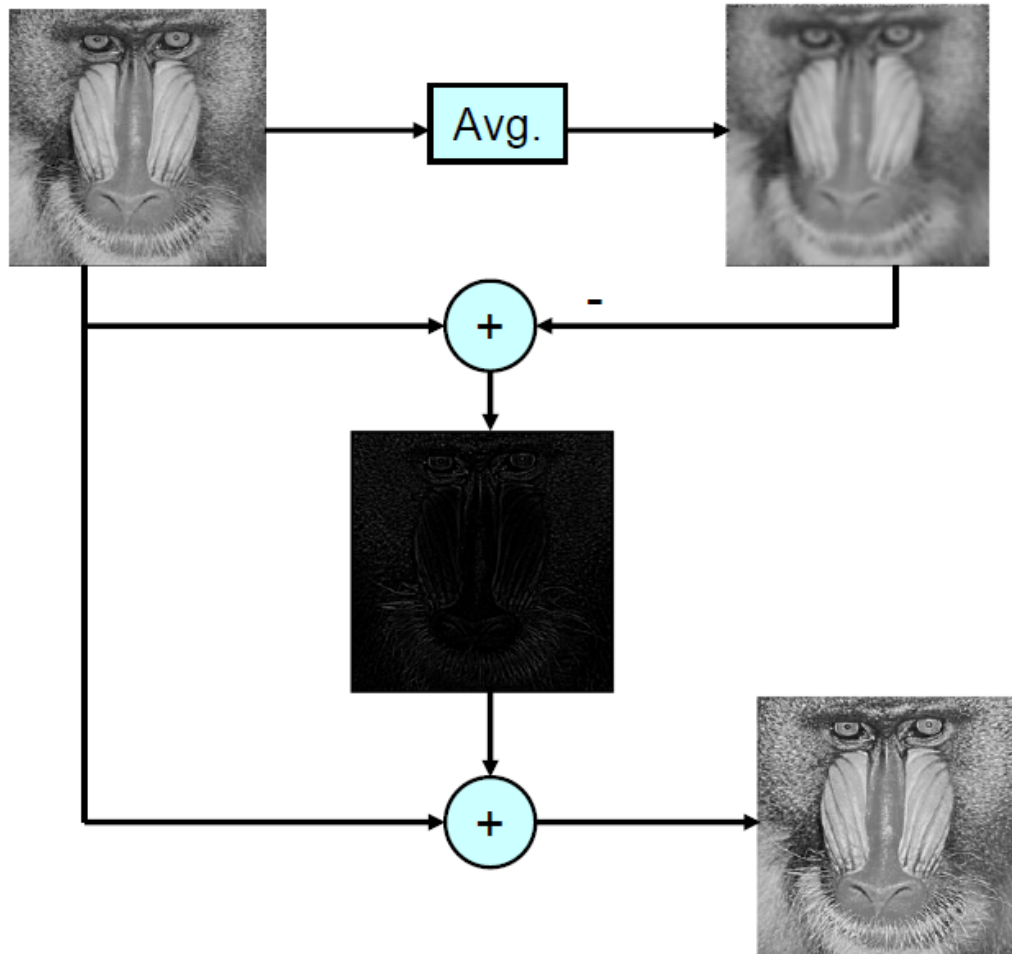
FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).



■ Steps

- Blur the image
- Subtract the blurred version from the original (this is called the *mask*)
- Add the "mask" to the original

High-boost filtering



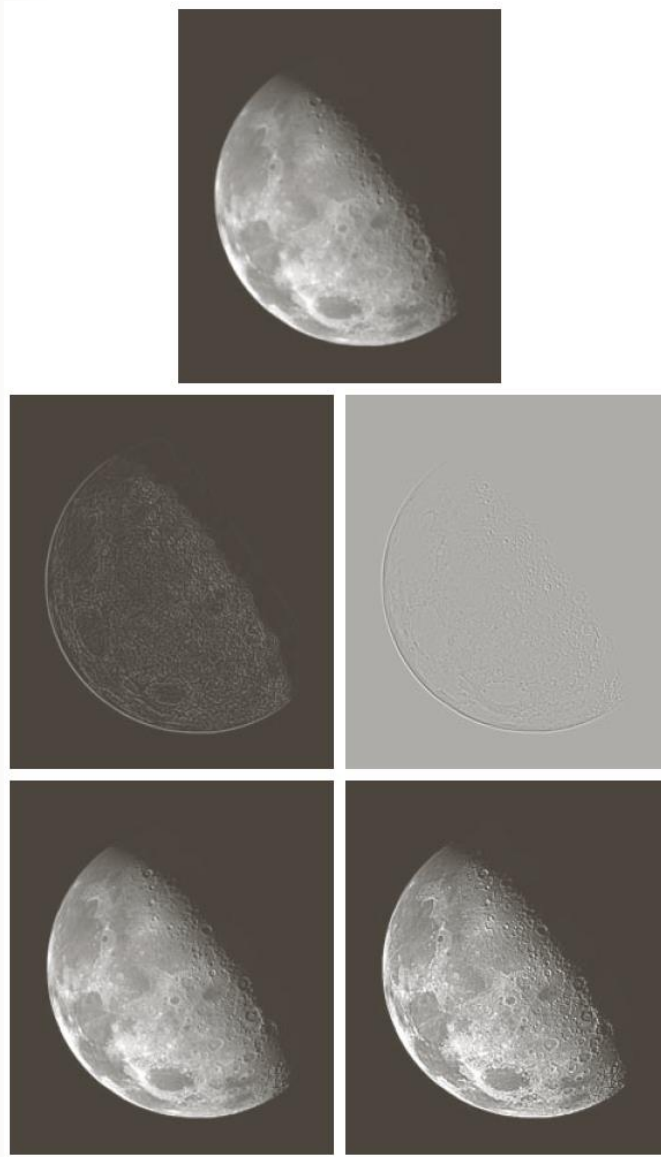
0	-1	0	-1	-1	-1
-1	$A + 4$	-1	-1	$A + 8$	-1
0	-1	0	-1	-1	-1

$$f_{hb}(x, y) = Af(x, y) - f_{lp}(x, y)$$

Unsharp mask:

high-boost with $A=1$

Unsharp mask example



a
b c
d e

FIGURE 3.38

(a) Blurred image of the North Pole of the moon.

(b) Laplacian without scaling.

(c) Laplacian with scaling.

(d) Image sharpened using the mask in Fig. 3.37(a).

(e) Result of using the mask in Fig. 3.37(b).

(Original image courtesy of NASA.)

Unsharp mask example

The Moon 25th August 2005 1:19 GMT

★ ADD TO FAVES
📄 BLOG THIS
🔍 ALL SIZES



Waning Gibbous Moon, 1:19 GMT Location Edinburgh, Scotland, UK
20.6 days old. Mirror Image!

The Moon with unsharp mask applied

★ ADD TO FAVES
📄 BLOG THIS
🔍 ALL SIZES



Similar to last nights picture but with some unsharp masking and turned into a greyscale picture. Do you think it helps?

Waning Gibbous Moon, 1:19 GMT Location Edinburgh, Scotland, UK
20.6 days old. Mirror Image! 25th August 2005 1:19 GMT

Unsharp mask example

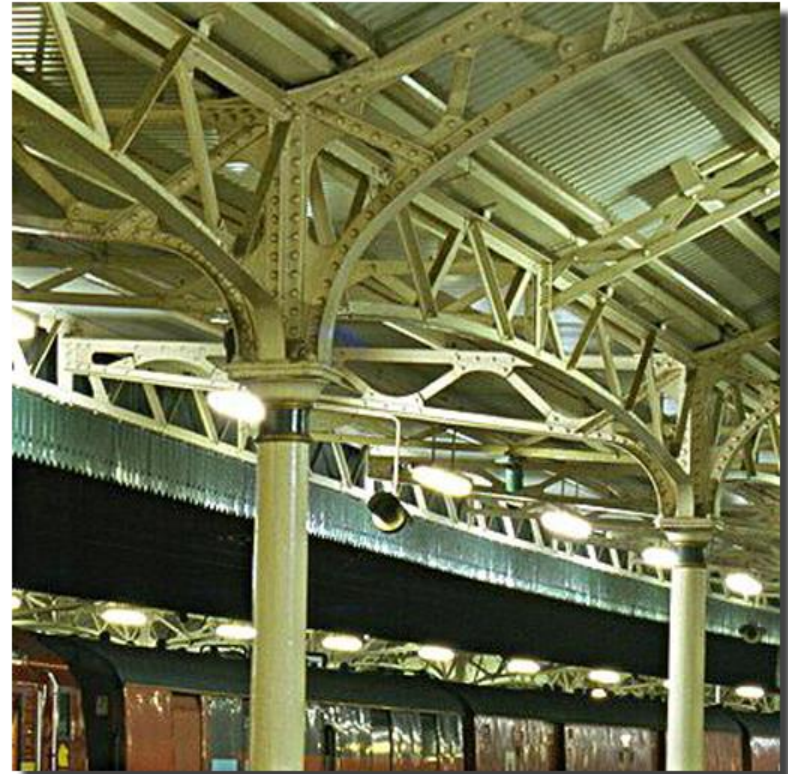
**Berries and melting ice after ice storm on
Feb 13th With Unsharp Mask**

☆ FID TO BLOG ALL
FIVES THIS SIZE

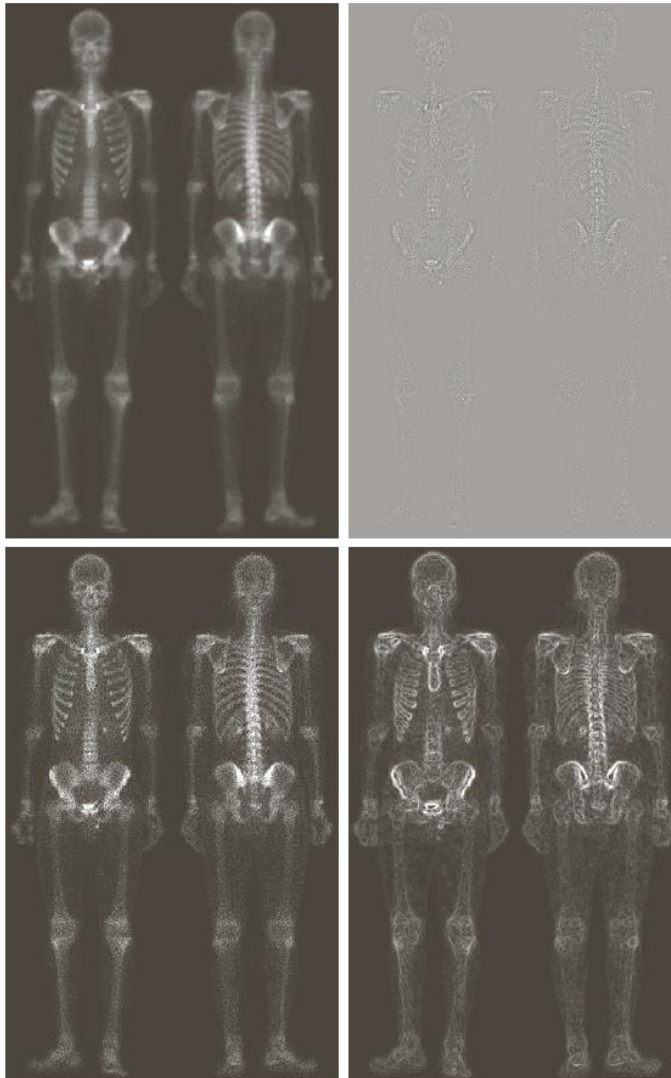


<http://flickr.com/photos/dpgnashua/2274968238/>

Unsharp mask example



Combining spatial operations



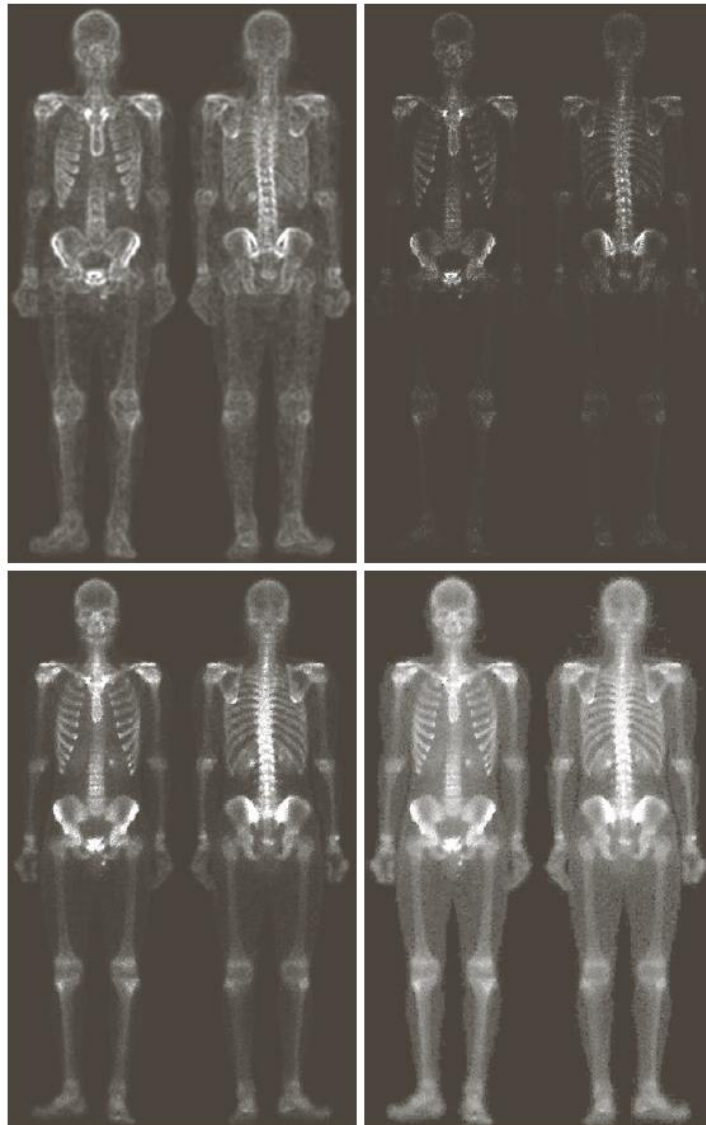
a	b
c	d

FIGURE 3.43

(a) Image of whole body bone scan.

(b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of (a).

Combining spatial operations



e f
g h

FIGURE 3.43

(Continued)

(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e).

(g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)



How to make good presentations

How to avoid a boring presentation?



"You're not allowed to use the sprinkler system to keep your audience awake."





Presentation Skills -- 1

- *Presentations are one of the first managerial skills which an engineer must acquire*
- **Formulate your Objectives**
- **Identify the Audience**
- **Structure**
 - **Beginning**
 - **Ending**
- **Make Good Slides**



Presentation Skills -- 2

- Be confident
- Keep eye contacts
- Appropriate posture (body language)
- Make sure all the audience can hear you
 - Speak slowly and clearly
 - Speak loudly
 - Pause
- Know all the technical terms you may be using. Also avoid using abbreviations or jargon terms that the audience will not understand.
- Smile
- **Practice**



Handling Difficult Questions

- Make sure you understand the question.
- If you are not able to answer a question then it is fine to say so!
- Ask if anyone else in the group knows the answer.
- Let the participant know that you will get back to them at a later date with the answer - and make sure you get back to them!



Dealing with questions

- It is **OK** to say “I don’t know” for the questions you don’t know the answer!!!
- Some phrases which can be useful when you want to avoid questions:
 - That’s not really my field, but I can put you in touch with someone who is an expert in the field.
 - Well, I think that **goes beyond the scope** of today’s presentation.
 - I’m afraid **we’ve run out of time**.
 - I haven't got the precise information with me today.
 - That's not really for me to say.
 - This is not really the place to discuss that matter.
 - Perhaps that's a question for another meeting.

Making PowerPoint Slides



Avoiding the Pitfalls of Bad Slides

<http://www.iasted.org/conferences/formatting/Presentations-Tips.ppt>

<https://www.assertion-evidence.com/>



Tips to be Covered

- Outlines of your presentation
- Slide Structure
- Fonts
- Color
- Background
- Graphs
- Spelling and Grammar
- Conclusions



Outline

- Make your 1st or 2nd slide an outline of your presentation
 - Ex: previous slide
- Follow the order of your outline for the rest of the presentation
- Only place main points on the outline slide
 - Ex: Use the titles of each slide as main points



Slide Structure – Good

- Use 1-2 slides per minute of your presentation
- Write in point form, not complete sentences
- Include 4-5 points per slide
- Avoid wordiness: use key words and phrases only



Slide Structure - Bad

- This page contains too many words for a presentation slide. It is not written in point form, making it difficult both for your audience to read and for you to present each point. Although there are exactly the same number of points on this slide as the previous slide, it looks much more complicated. In short, your audience will spend too much time trying to read this paragraph instead of listening to you.



Slide Structure – Good

- Show one point at a time:
 - Will help audience concentrate on what you are saying
 - Will prevent audience from reading ahead
 - Will help you keep your presentation focused



Slide Structure - Bad

- Do not use distracting animation
- Do not go overboard with the animation
- Be consistent with the animation that you use



Fonts - Good

- Use at least an 18-point font
- Use different size fonts for main points and secondary points
 - this font is 24-point, the main point font is 28-point, and the title font is 36-point
- Use a standard font like Times New Roman or Arial



Fonts - Bad

- If you use a small font, your audience won't be able to read what you have written
- CAPITALIZE ONLY WHEN NECESSARY. IT IS DIFFICULT TO READ
- **Don't use a complicated font**



Color - Good

- Use a color of font that contrasts sharply with the background
 - Ex: blue font on white background
- Use color to reinforce the logic of your structure
 - Ex: light blue title and dark blue text
- Use color to emphasize a point
 - But only use this occasionally



Colour - Bad

- Using a font colour that does not contrast with the background colour is hard to read
- Using colour for decoration is distracting and annoying.
- Using a different colour for each point is unnecessary
 - Using a different colour for secondary points is also unnecessary
- Trying to be creative can also be bad



Background - Good

- Use backgrounds such as this one that are attractive but simple
- Use backgrounds which are light
- Use the same background consistently throughout your presentation

Background – Bad

- Avoid backgrounds that are distracting or difficult to read from
- Always be consistent with the background that you use





Graphs - Good

- Use graphs rather than just charts and words
 - Data in graphs is easier to comprehend & retain than is raw data
 - Trends are easier to visualize in graph form
- Always title your graphs

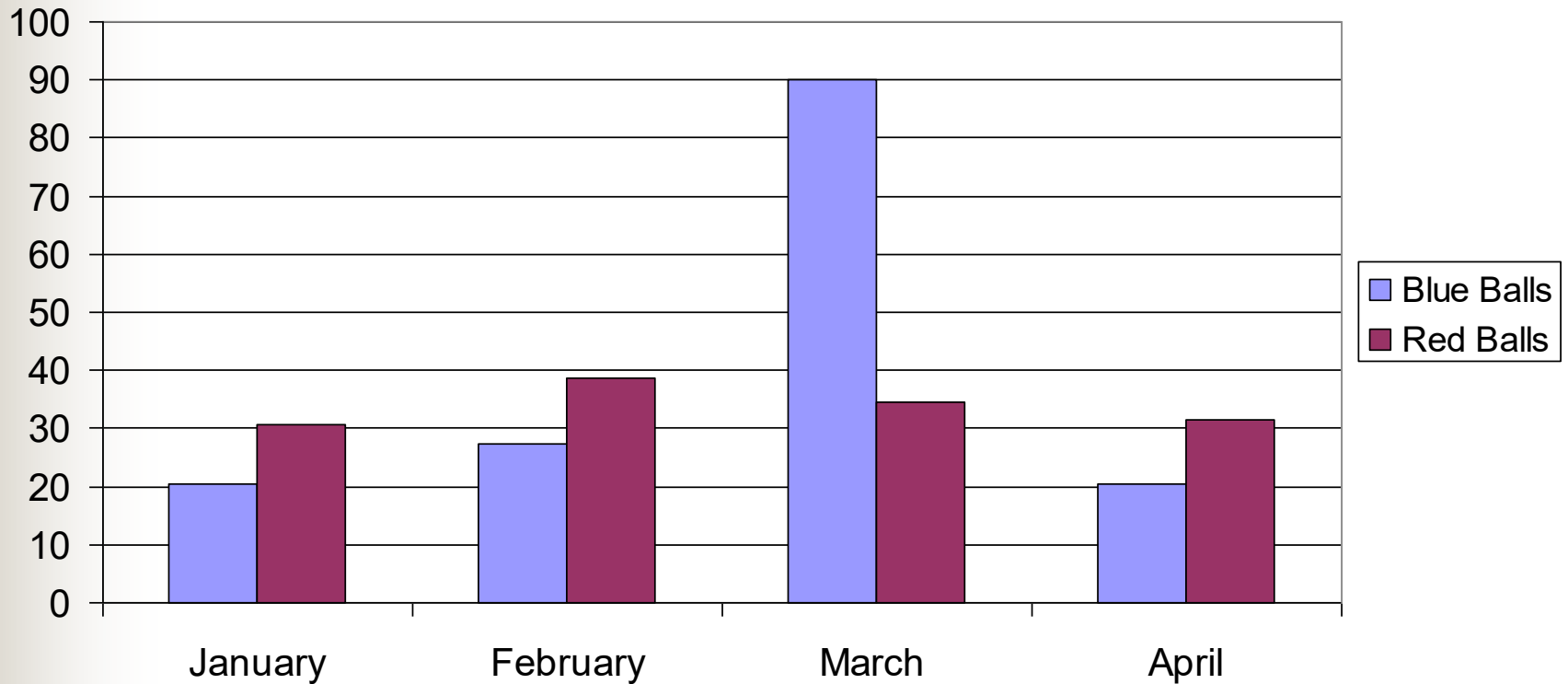


Graphs - Bad

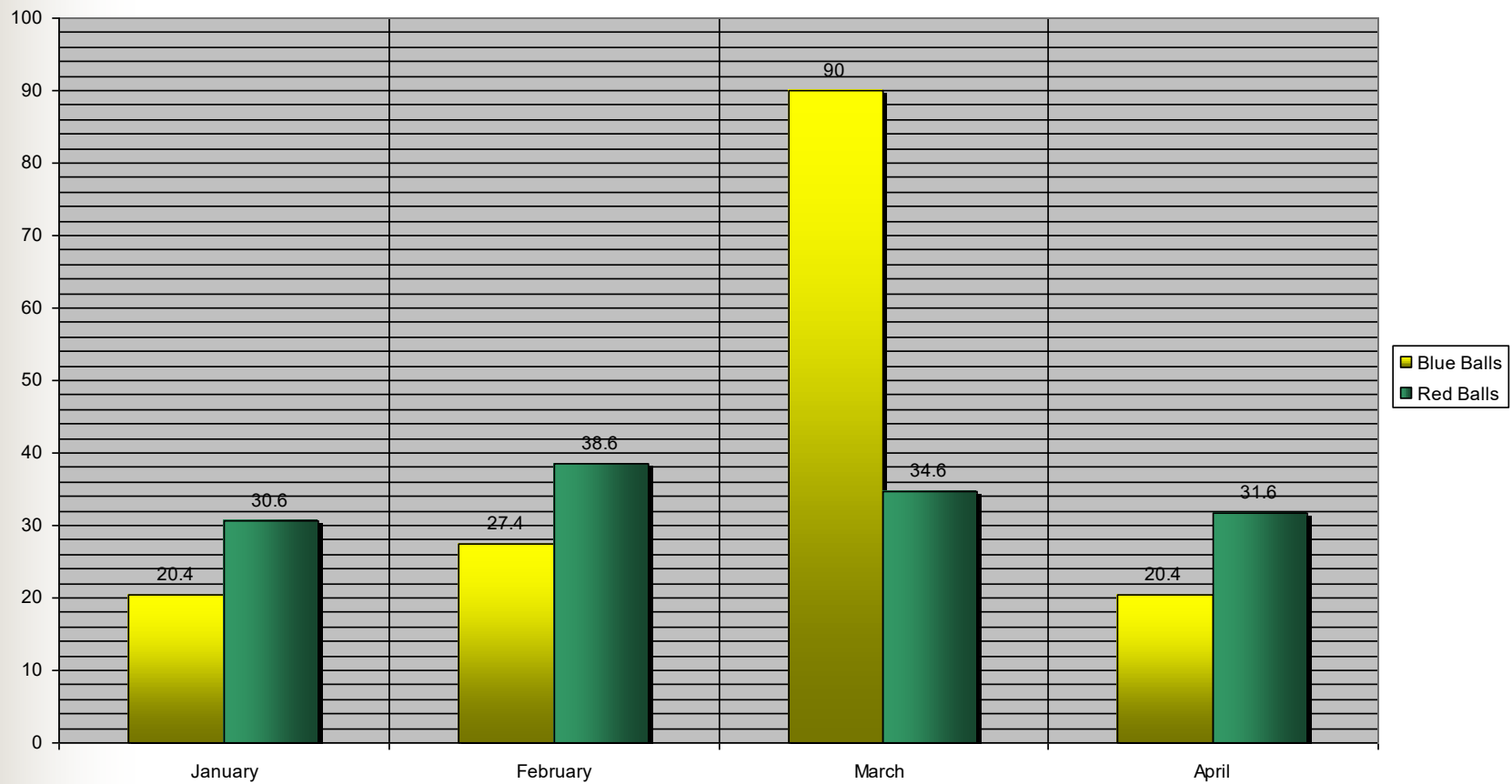
	January	February	March	April
Blue Balls	20.4	27.4	90	20.4
Red Balls	30.6	38.6	34.6	31.6

Graphs - Good

Items Sold in First Quarter of 2002



Graphs - Bad





Graphs - Bad

- Minor gridlines are unnecessary
- Font is too small
- Colours are illogical
- Title is missing
- Shading is distracting



Spelling and Grammar

- Proof your slides for:
 - spelling mistakes
 - the use of of repeated words
 - grammatical errors you might have make

- If English is not your first language, please have someone else check your presentation!



Conclusion

- Use an effective and strong closing
 - Your audience is likely to remember your last words

- Use a conclusion slide to:
 - Summarize the main points of your presentation
 - Suggest future avenues of research



Improving your presentation skills

- Practice
 - Practice
 - Practice
 - Recording your presentations
-
- The **content of a presentation** is just as important as its format.



Announcement

- Roy will present HW1 today.
- HW2 is out today, due: 3/3.
- Ali will present HW2 on 3/4.



HW 1 PRESENTATION